



# Module 2 – DDL (Data Definition Language)

A professional training session for fresh engineering graduates. Duration: 45 minutes | Prerequisites: Basic Database Concepts & Module 1 Completed



# What You'll Learn Today

By the end of this module, you will be equipped to define and manage database structures using DDL commands.

## Database & Schema

Understand logical containers and organization of database objects.

## Tables & Data Types

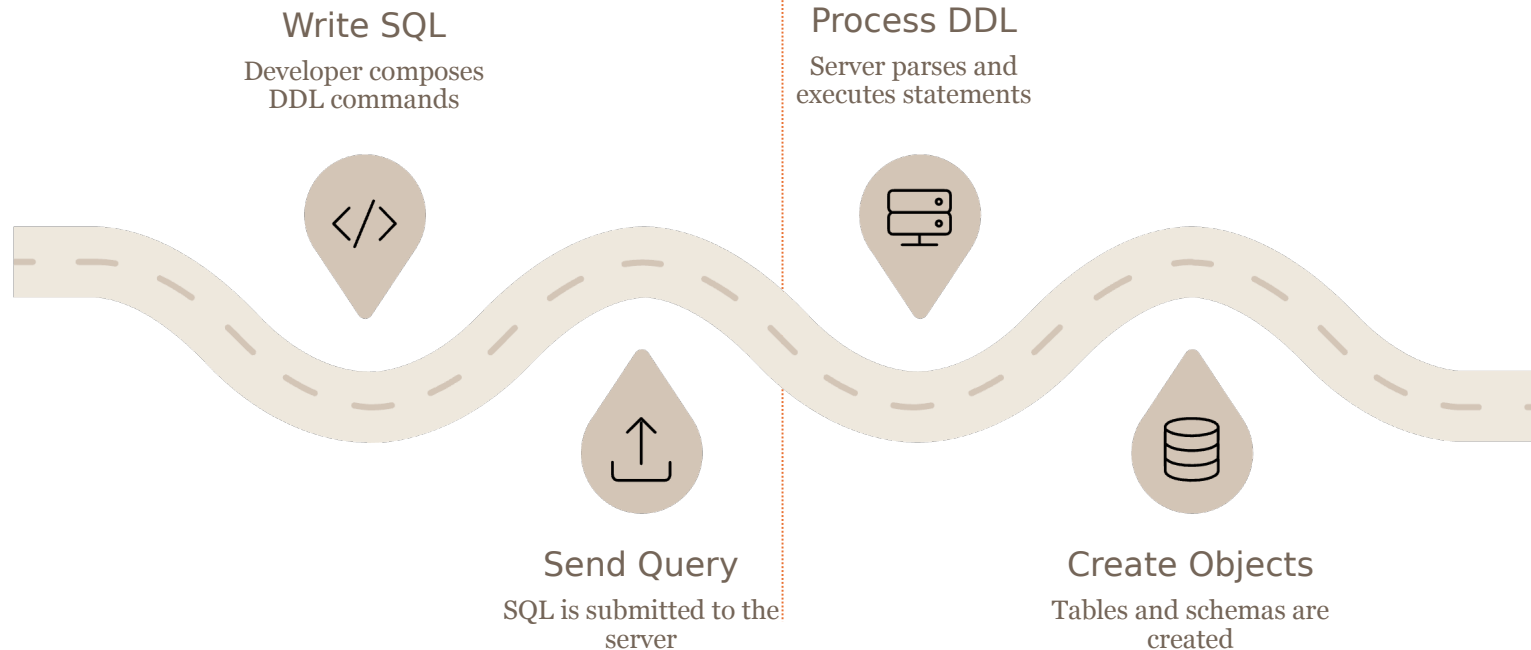
Learn table structure and common MySQL data types.

## DDL Commands

CREATE, ALTER, DROP, TRUNCATE, and RENAME — in depth.

## Key Takeaway

DDL defines and manages the *structure* of a database, not its data.





# What Is a Database?

A **database** is a logical container that stores related data in an organized, accessible way. Think of it as a digital filing system for an entire institution.

## Example: CollegeDB

- Students
- Faculty
- Courses
- Marks

# Schema & Table Structure

## Schema

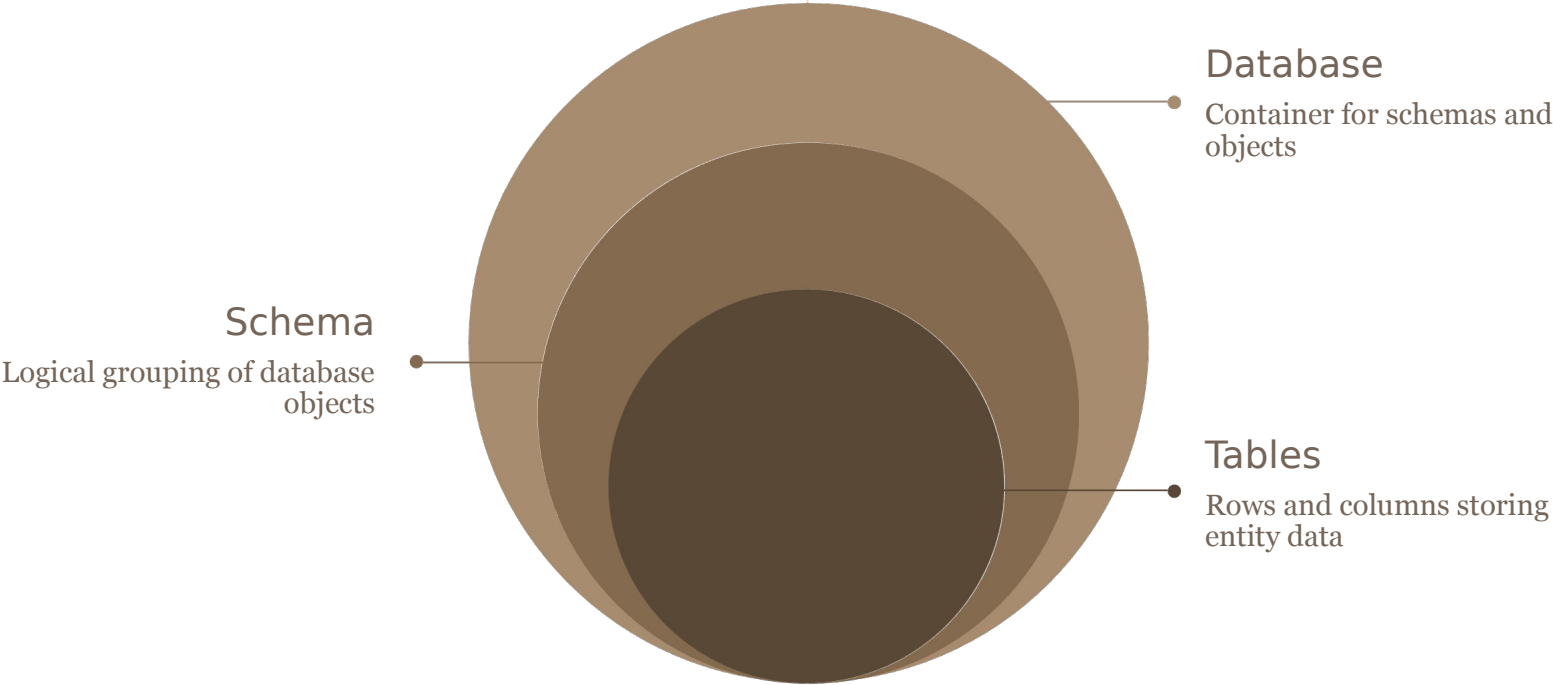
A schema is a logical organization of database objects within a database.

- Tables
- Views
- Procedures
- Functions
- Indexes

## Table

A table stores information in **rows** (records) and **columns** (fields), representing one entity.

| Student_ID | Name   | Branch |
|------------|--------|--------|
| 101        | Rahul  | CSE    |
| 102        | Anjali | ECE    |



# Understanding Tables

A table represents **one entity** in your database. Every table is made up of rows (records) and columns (fields).



## Employee Table

Stores employee records including ID, name, and salary.



## Customer Table

Tracks customer information and contact details.



## Orders Table

Records all purchase orders and transaction data.



## Products Table

Maintains product catalog with pricing and inventory.

| Employee_ID | Name  | Salary |
|-------------|-------|--------|
| 101         | John  | 50,000 |
| 102         | David | 60,000 |

# MySQL Data Types

Every column in a table **must** have a defined data type. Choosing the right type ensures data integrity and storage efficiency.

## INT

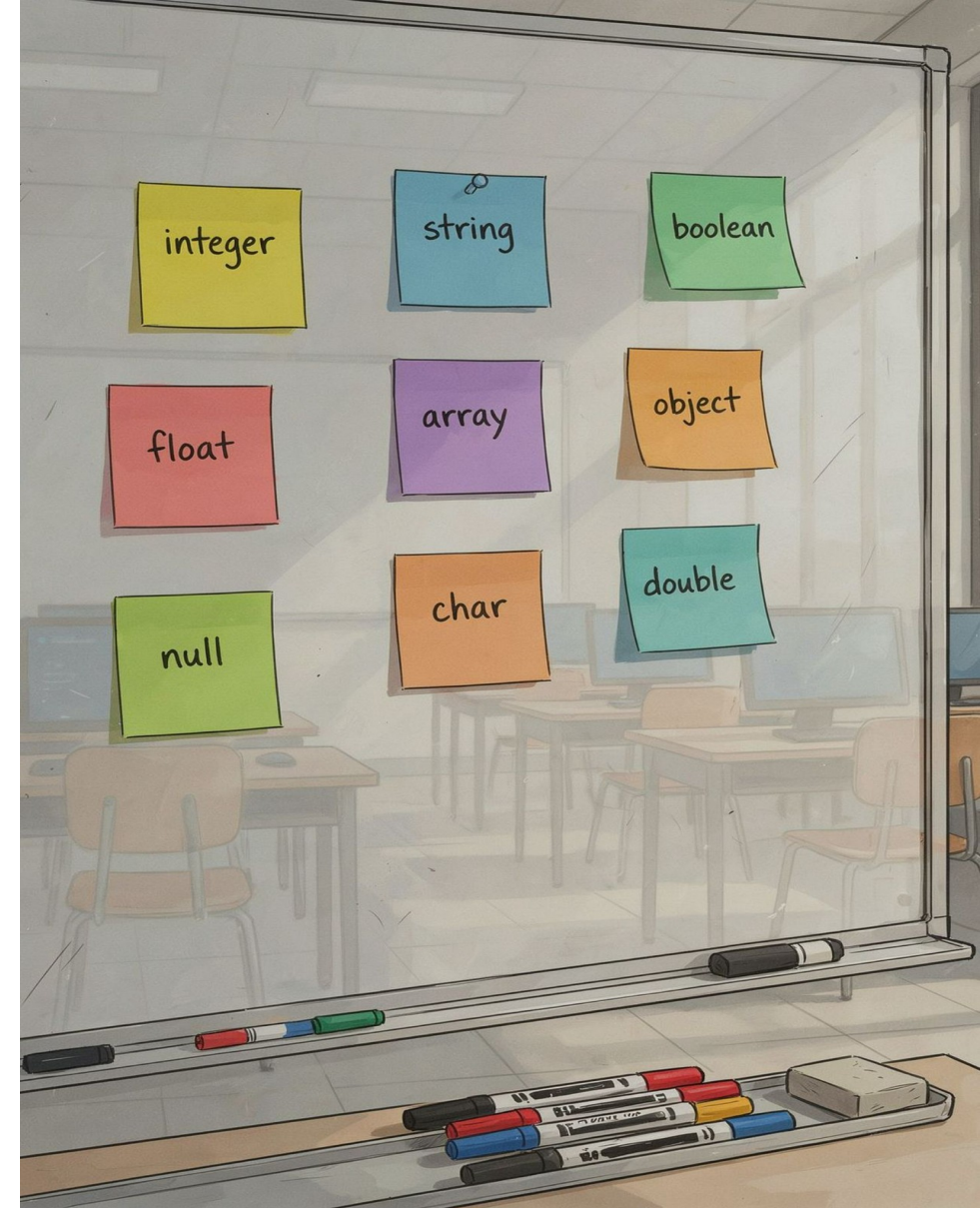
Stores whole numbers. Example: 101, 2500, 99999

## VARCHAR

Stores variable-length text. Example: Rahul, Bangalore, Cloud Engineer

## DATE

Stores only a date value. Example: 2026-08-03



# More MySQL Data Types

## DATETIME

Stores date and time together.

2026-08-03 14:30:20

## DECIMAL

Stores precise decimal values for financial data.

45000.75

## BOOLEAN

Stores logical true/false values.

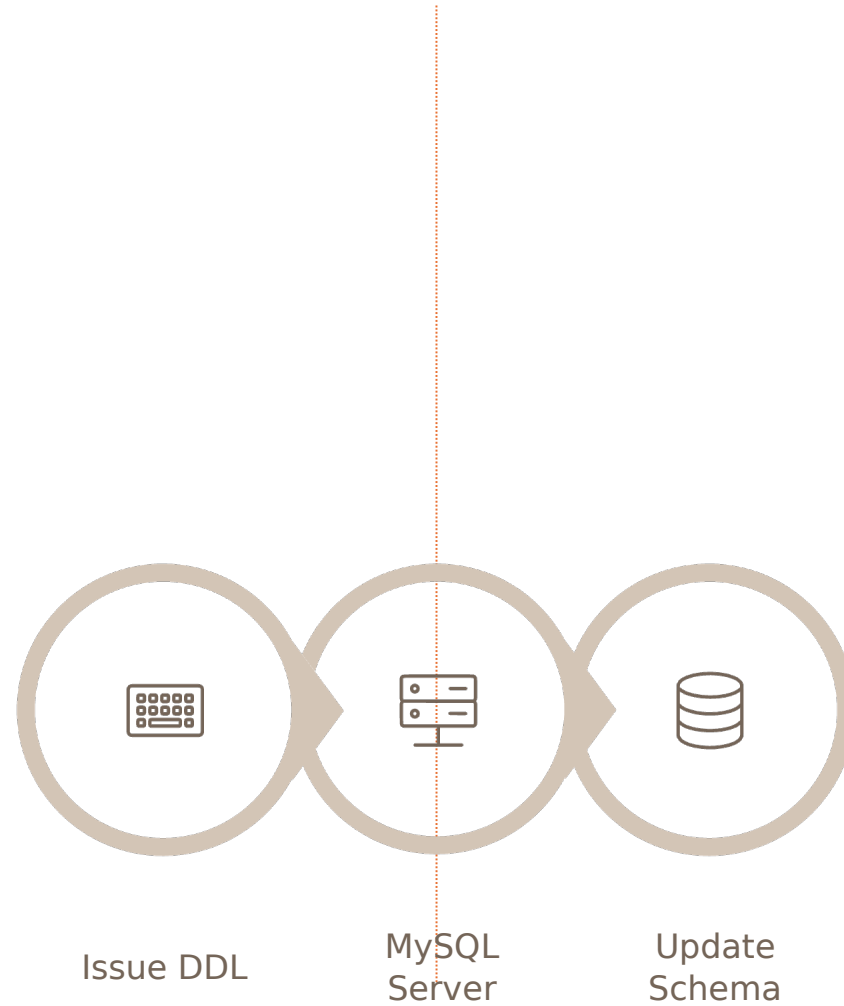
TRUE / FALSE

📌 Always choose the most specific data type for each column — it improves performance and prevents invalid data entry.

# Introduction to DDL

**DDL - Data Definition Language** is used to define and modify the *structure* of a database. DDL changes the schema, not the data itself.

- CREATE
- ALTER
- DROP
- TRUNCATE
- RENAME





# CREATE Command

The **CREATE** command is used to build new database objects — databases, tables, indexes, and views — from scratch.

## Create a Database

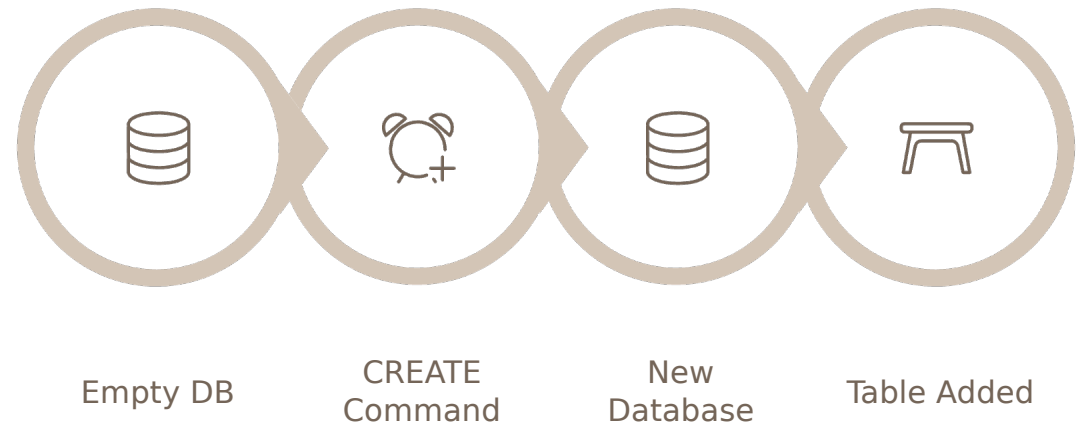
```
CREATE DATABASE CollegeDB;
```

## Create a Table

```
CREATE TABLE Students (  
  Student_ID INT,  
  Name VARCHAR(100),  
  DOB DATE  
);
```

## What You Can CREATE

-  Database
-  Tables
-  Indexes
-  Views



# ALTER Command

The **ALTER** command modifies the structure of an existing database object — without deleting any existing data.

## Common Uses

- Add a Column
- Modify a Column
- Drop a Column
- Rename a Column

## Add a Column

```
ALTER TABLE Students
ADD Email VARCHAR(100);
```

## Modify a Column

```
ALTER TABLE Students
MODIFY Name VARCHAR(150);
```

### Before Table

Original columns:  
ID, Name, Age

### Updated Table

New columns: ID,  
Name, Age, Email

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### ALTER Applied

```
ALTER TABLE
Students ADD Email
VARCHAR(100)
```

# DROP Command

The **DROP** command completely removes a database object — along with *all of its data*. This action is permanent and cannot be undone.

## Drop a Table

```
DROP TABLE Students;
```

## Drop a Database

```
DROP DATABASE CollegeDB;
```

### ⚠ Critical Warning

DROP permanently deletes **both the object and all its data**. Always back up before executing DROP in production environments.

**Table Exists**  
A table with stored data is present.

**Permanently Deleted**  
Object and all data are removed forever.

**DROP Executed**  
The DROP command is run against the object.

# TRUNCATE Command

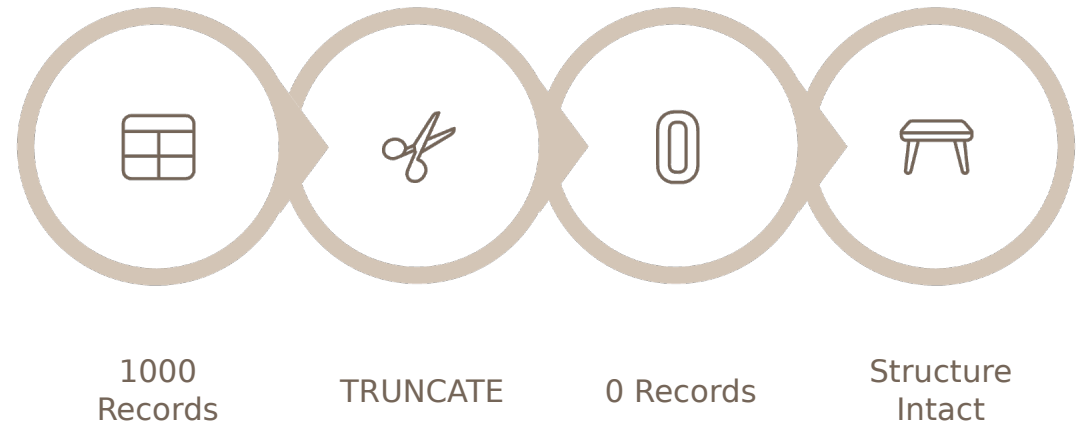
The **TRUNCATE** command deletes *all records* from a table while keeping the table structure intact — faster and more efficient than DELETE.

## Syntax

```
TRUNCATE TABLE Students;
```

## Benefits

-  Faster than DELETE
-  Keeps table definition
-  Resets storage efficiently



# RENAME Command

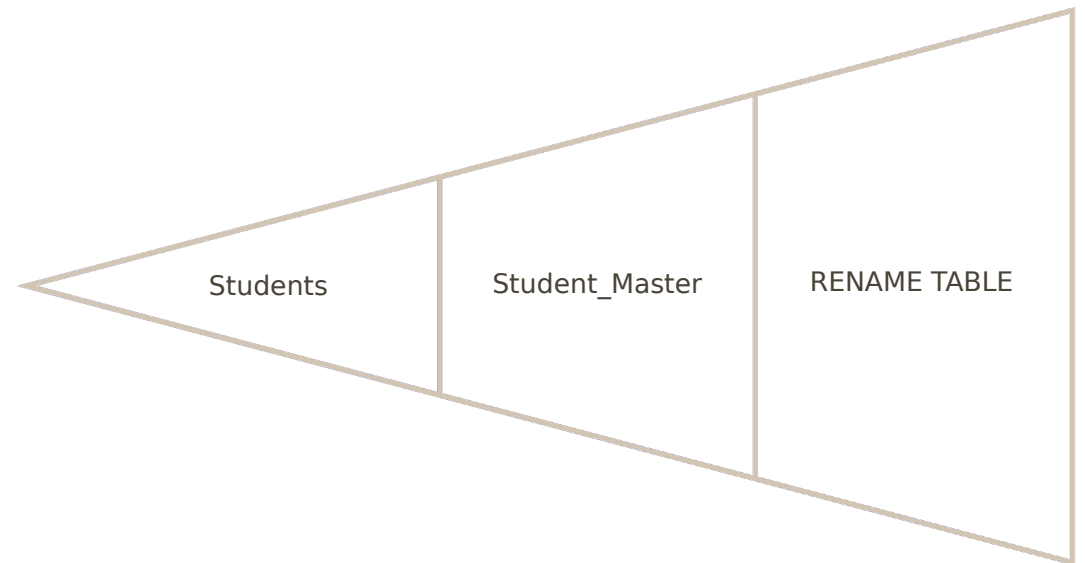
The **RENAME** command changes the name of an existing table. No data is lost and the structure remains unchanged.

## Syntax

```
RENAME TABLE Students  
TO Student_Master;
```

## When to Use RENAME

- Refactoring database design
- Naming standardization
- Project migration



# DDL Command Comparison

A quick reference to understand what each DDL command does to your database structure and data.

| Command  | Purpose             | Data Removed? | Structure Changed? |
|----------|---------------------|---------------|--------------------|
| CREATE   | Creates objects     | No            | Yes                |
| ALTER    | Modifies objects    | No            | Yes                |
| DROP     | Deletes object      | Yes           | Yes                |
| TRUNCATE | Deletes all rows    | Yes           | No                 |
| RENAME   | Changes object name | No            | Yes                |

|                  |                 |                |
|------------------|-----------------|----------------|
| CREATE → Build   | ALTER → Modify  | DROP → Destroy |
| TRUNCATE → Empty | RENAME → Rename |                |

# What We Learned

01

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## Database & Schema

A database is a logical container; a schema organizes objects like tables, views, and procedures within it.

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## MySQL Data Types

INT, VARCHAR, DATE, DATETIME, DECIMAL, and BOOLEAN — each serving a specific storage purpose.

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## Tables & Structure

Tables represent entities, organized into rows (records) and columns (fields).

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## DDL Commands

CREATE, ALTER, DROP, TRUNCATE, and RENAME — the five commands that define database structure.

"A well-designed database starts with a well-defined structure, and DDL is the language that builds it."

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Module 2 – Data Definition Language | 45-Minute Instructor-Led Session





**Thank you**

You're Ready to  
Build Databases

